Chai: supplementary material

to "Chai: A Signed Three-Valued Algebra from Parameterized Boolean Negation"

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This file is generated by `make_supplement.py` from the results of `chai_checker.py --full` and the other scripts of the package; it is not edited by hand, and the generator stops if any result file lacks data. Notation as in the paper; $-,0,+$ stand for $-1,0,+1$. The main paper contains the definitions, truth tables, and proofs of the arbitrary-arity generation theorems and of the completeness classification without primitive nullary constants. This supplement provides detailed finite closure data, additional term catalogues, and the computational evidence for the classification with nullary constants. Sections S0–S19 are cited from the paper as "supplement Sn"; S14–S16 (quarter turn, complex table, folding, sign rule) and the constants case of S6 are checked by `verify_direction_rotation.py`; S17 reproduces the XY3 and G3 tables of the published divisional from their definitions, with hexaogram and bi-trigram symbols; S18 lists the checks added for this version (`chai_verify_v1_increments.py`); S19 reproduces the structure diagram of the earlier version.

S0. What follows from classical theory and what is added here

The table lists each result of the paper, the classical statement it rests on, and what the paper adds. The classical sources are Post (1921), Słupecki (1939), Rosenberg (1970), Lau (2006) and, for strong Kleene logic, Robles and Méndez (2022). The comparison is limited to the sources cited in the paper.

result	classical background	added here
Minimal carrier and the three sections (Prop. 1)	The least subset closed under an operation and containing a given set is a standard construction.	That $\{0,1\}$ closed under $\mathsf{DIF}(a,b)=\operatorname{sgn}(a-b)$ gives exactly $\{-1,0,+1\}$, relative to the chosen saturation; the three sections of DIF at the constants and the fact that only ANT is an involution. Elementary.
Boolean layer (Thm. 5)	The functions mapping B^n into B form a clone (an invariant-relation clone; Rosenberg, Lau).	SYN, OR, XOR, AND, NOT generate *exactly* this clone, with explicit terms, and with arbitrary values at the points containing -1.
Zero-fixing barrier (Thm. 3)	$\mathscr{Z}=\mathrm{Pol}\{0\}$ is a maximal clone (Rosenberg; Lau). Hence any set inside $\mathscr{Z}$ is incomplete, and one operation outside $\mathscr{Z}$ can complete a set that generates $\mathscr{Z}$.	Explicit terms showing that the five binary operations generate *all* of $\mathscr{Z}_n$ (not only a subclone), and that one zero-breaking operation then gives every operation (Thm. 3c reproves maximality with terms).
Odd operations (Thm. 4)	The functions commuting with $x\mapsto -x$ (self-dual, the centralizer of ANT) form a clone; standard.	DIF alone generates this clone, with explicit terms k_c and s_p for every arity.
Classification, 40 of 127 (Thm. 6)	Post's and Słupecki's completeness criteria; precomplete classes.	Exact condition for this seven-operation signature, minimal sets, counts with and without nullary constants, checked subset by subset by two methods (S6).
Polarization $\mathsf{AND}=\mathsf{DIF}(\mathsf{OR},\mathsf{XOR})$ (Thm. 1)	Elementary inclusion–exclusion on $\{0,1\}$-valued functions.	The identity at the sign level on D, with the parity lemma, and its failure on $\{-m,\dots,m\}$, $m\geq 2$.
Involution $\Phi=(\mathsf{SYN},\mathsf{DIF})$ (Thm. 2)	None specific.	Finite statement for this pair of operations; elementary.
Sign rule of AND (Remark 1, S16)	$ab=\frac{1}{4}\big((a+b)^2-(a-b)^2\big)$, so $\operatorname{sgn}(ab)=\operatorname{sgn}(\lvert a+b\rvert-\lvert a-b\rvert)$ on all integers (quarter-square multiplication).	Only that on D the squares can be folded to signs before the comparison (Thm. 1); not a general principle (fails on $\{-2,\dots,2\}$).

result	classical background	added here
Constants as nullary operations, 95 of 127 (Remark 3, S6)	The Post–Słupecki criteria with constants; monotone operations and the relation ρ as invariants.	A short constructive proof for this signature (terms for NOT and ANT from SYN with one further operation) and the exact criterion; checked against the 127 subsets by a separate script.
Quarter turn of the truth table (Remark 2, S14)	Rotating a matrix is precomposition of the function with a rotation of the argument grid; multiplication by i is the quarter turn of the complex plane.	That for Chai the turn carries SYN $\to$ DIF $\to -$SYN, OR $\leftrightarrow$ XOR, AND $\to -$AND; the link with the order-8 symmetry group and the parity of functions. Elementary.
Complex table SYN $+\, i$ DIF (supplement only, S15)	sgn applied to both coordinates commutes with multiplication by i (the dihedral symmetry of $\mathbb{Z}^2$).	The identification of the two tables with the saturated real and imaginary parts of $(1+i)\overline{(a+ib)}$, and with the sum and difference views of a BID. Elementary.
Folding is not a quotient (Prop. 5); simplicity and the rest of S16 (supplement only)	A functionally complete (primal) finite algebra with constants has no nontrivial congruence, subalgebra or automorphism; standard.	The exact list of the operations that respect ker MAG (only AND, ANT), the folded complex table, and the Boolean/set readings. Elementary; the set statements are classical.
K_3, L_3 as terms (Prop. 3, S3)	Completeness implies that every function is a term.	Explicit minimal terms of familiar connectives, found by exhaustive search (S3, S5).
Difference lift, orbit count (Section VI, Prop. 6 and 7)	Elementary counting and Burnside's lemma.	The counts for this setting. The XY3/G3/C9 layers themselves are described in the published divisional CN 122173050 A and are background.

S1. Algebraic laws of the seven binary operations

Associativity failures are counted over the 27 ordered triples.

operation	commutative	assoc. failures	idempotent	identity	absorbing
SYN	yes	4/27	yes	0	–
DIF	no	16/27	no	–	–
OR	yes	8/27	no	–	–
XOR	yes	8/27	no	–	–
AND	yes	0/27	no	+	0
MAX	yes	0/27	yes	-	+
MIN	yes	0/27	yes	+	-

Left-distributive pairs (f,g), meaning $f(a,g(b,c)) = g(f(a,b),f(a,c))$ for all a,b,c, among these seven operations: (SYN, MAX), (SYN, MIN), (AND, SYN), (AND, DIF), (MAX, MAX), (MAX, MIN), (MIN, MAX), (MIN, MIN).

S2. The pseudocomplement NOT

On $\{0,+1\}$, $\mathsf{NOT}(x) = 1 - x$ is the Boolean complement. On D, $\mathsf{NOT}(a) = \min\{x \in \{0,+1\} : \mathsf{MAX}(a,x) = +1\}$: it is the least Boolean element that joins a to $+1$, so $\mathsf{NOT}(-1) = \mathsf{NOT}(0) = +1$ and $\mathsf{NOT}(+1) = 0$. It is a *relative* dual pseudocomplement (relative to the sublattice $\{0,+1\}$), not a pseudocomplement of D, and $\mathsf{NOT} \circ \mathsf{NOT} = [0,0,+1] \neq \mathrm{id}$.

S3. The explicit terms with their complete tables

$$\delta_{+1}(x) = \mathsf{SYN}(x, \mathsf{OR}(x,x)), \qquad \delta_{-1}(x) = \mathsf{XOR}(x, \mathsf{OR}(x,x)),$$
$$h_{\pm 1}(u,v) = \mathsf{AND}(\delta_{+1}(u), \delta_{\pm 1}(v)), \qquad h_0(u,v) = \mathsf{DIF}(\delta_{+1}(u), \mathsf{AND}(\delta_{+1}(u), \mathsf{OR}(v,v))),$$
$$k_{-1}(u,v) = \mathsf{DIF}(u, \mathsf{DIF}(v, \mathsf{DIF}(v,u))), \qquad k_0(u,v) = \mathsf{DIF}(v, \mathsf{DIF}(v, \mathsf{DIF}(u,v))),$$
$$k_{+1}(u,v) = \mathsf{DIF}(\mathsf{DIF}(u,v), \mathsf{DIF}(\mathsf{DIF}(u,v),v)), \qquad \mathsf{MAG}(x) = \mathsf{NOT}(\mathsf{DIF}(x, \mathsf{DIF}(x, \mathsf{NOT}(x)))).$$

Unary terms ($\delta_{+1} = [x{=}+1]$, $\delta_{-1} = [x{=}-1]$, $\mathsf{MAG}(x) = x^2$):

x	$-$	0	$+$
δ_{+1}	0	0	$+$
δ_{-1}	$+$	0	0
MAG	$+$	0	$+$

Lemma 2: h_{+1}, h_{-1}, h_0 equal $[u{=}{+1}][v{=}{+1}]$, $[u{=}{+1}][v{=}{-1}]$, $[u{=}{+1}][v{=}0]$ (rows u, columns v):

$h_{-1}(u,v)$	$-$	0	$+$
$-$	0	0	0
0	0	0	0
$+$	$+$	0	0

$h_0(u,v)$	$-$	0	$+$
$-$	0	0	0
0	0	0	0
$+$	0	$+$	0

$h_{+1}(u,v)$	$-$	0	$+$
$-$	0	0	0
0	0	0	0
$+$	0	0	$+$

Theorem 4: $k_c(u,v) = u$ if $u \neq 0$ and $v = cu$, and 0 otherwise (so k_{-1} is u exactly where $v = -u$, k_0 where $v = 0$, k_{+1} where $v = u$):

$k_{-1}(u,v)$	$-$	0	$+$
$-$	0	0	$-$
0	0	0	0
$+$	$+$	0	0

$k_0(u,v)$	$-$	0	$+$
$-$	0	$-$	0
0	0	0	0
$+$	0	$+$	0

$k_{+1}(u,v)$	$-$	0	$+$
$-$	$-$	0	0
0	0	0	0
$+$	0	0	$+$

All of these tables are asserted by the generator against the intended functions at every point of D or D^2.

Familiar binary connectives as terms (moved here from the paper; K_3 and L_3 coding $-1 = F$, $0 = U$, $+1 = T$). Size: number of operator nodes of the term (tree size, repeated subterms counted each time), equal to the minimum over the seven operators found by exhaustive search up to size 5; each term below is evaluated by the generator at all nine points and compared with the stated function.

connective	meaning	size	term
EQ	$[a = b]$	2	`NOT(XOR(a,b))`
LE	$[a \leq b]$	2	`NOT(DIF(a,b))`
LT	$[a < b]$	3	`NOT(NOT(DIF(b,a)))`
MAX	$\max(a,b)$	3	`SYN(SYN(a,b),XOR(a,b))`
MIN	$\min(a,b)$	3	`DIF(SYN(a,b),XOR(a,b))`
imp_{K_3}	$\max(-a,b)$	3	`SYN(DIF(b,a),OR(a,b))`
imp_{L_3}	$\min(1,1-a+b)$	4	`SYN(NOT(a),SYN(b,AND(a,b)))`
and_{L_3}	$\max(-1,a+b-1)$	4	`DIF(SYN(a,b),NOT(AND(a,b)))`
or_{L_3}	$\min(1,a+b+1)$	4	`SYN(SYN(a,b),NOT(AND(a,b)))`
CARRY	half-adder carry	3	`SYN(a,DIF(b,SYN(a,b)))`
SUM	$a+b \bmod 3$	5	`AND(SYN(a,b),DIF(XOR(a,b),AND(a,b)))`

The standard connectives are terms, not primitives, and the primitives differ from them: $\text{NOT}(0) = +1 \neq \text{ANT}(0) = 0$, $\text{AND}(-1,-1) = +1 \neq \min(-1,-1) = -1$, and OR is not the lattice join. The K_3 and L_3 implications differ only at $(0,0)$ ($U \to U$ is U in K_3 and T in L_3).

S4. Generation heights

Closure of the binary slice D^{D^2}: number of new functions at heights $0, 1, 2, \ldots$ (the two projections, and in the last six rows also the three constants, are at height 0). Produced by `chai_checker.py --full` and `--full-consts`; the logs are `full_run.log` and `consts_*.log`.

generators	functions	new per height
SYN, DIF, OR, XOR, AND (zero-fixing clone $\mathscr{Z}_2$)	6561	2, 9, 143, 3811, 2596
DIF, NOT	19683	2, 5, 35, 353, 5030, 13486, 772
SYN, ANT, NOT	19683	2, 5, 24, 225, 2622, 15160, 1645
SYN, OR, XOR, AND, NOT (M_1, incomplete)	3888	2, 9, 106, 1494, 2275, 2
ANT, OR, XOR, AND, NOT (M_2, incomplete)	1225	2, 10, 89, 792, 332
DIF + nullary constants	19683	5, 12, 78, 1092, 11656, 6838, 2
SYN, OR + nullary constants	19683	5, 10, 83, 1248, 11703, 6634
SYN, XOR + nullary constants	19683	5, 10, 97, 1645, 15198, 2728
SYN, AND + nullary constants	19683	5, 10, 103, 3000, 16086, 479
SYN, NOT + nullary constants	19683	5, 7, 38, 335, 3693, 13731, 1874
SYN, ANT + nullary constants	19683	5, 7, 37, 425, 5466, 13693, 50

The closure of DIF alone without constants is the set of $81 = 3^4$ odd binary functions. The first row has $6561 = 3^8$ functions, the binary slice of $\mathscr{Z}$; the two rows for M_1 and M_2 stay below 19683, as the classification requires.

S5. The 27 unary functions as terms

Code: digits $0,1,2$ for $0,+1,-1$, listing $g(-1), g(0), g(+1)$, prefix `0t`. Size: minimum number of operator nodes (tree size, repeated subterms counted each time; exhaustive search by size); term: the lexicographically smallest minimal term. Three signatures: all seven operations (the table of v9, Table IV; maximum size 4), $\{\mathsf{DIF}, \mathsf{NOT}\}$ (maximum 8) and $\{\mathsf{SYN}, \mathsf{ANT}, \mathsf{NOT}\}$ (maximum 11). The three odd unary functions 0, x, $\mathsf{ANT}(x)$ are generated by DIF alone: $\mathsf{DIF}(x,x)$, x, $\mathsf{DIF}(\mathsf{DIF}(x,x),x)$.

code	g	size	all seven	size	{DIF,NOT}	size	{SYN,ANT,NOT}
0t000	$[0,0,0]$	1	DIF(a,a)	1	DIF(a,a)	2	SYN(ANT(a),a)
0t001	$[0,0,+]$	2	NOT(NOT(a))	2	NOT(NOT(a))	2	NOT(NOT(a))
0t002	$[0,0,-]$	3	ANT(NOT(NOT(a)))	4	DIF(DIF(NOT(a),a),NOT(a))	3	ANT(NOT(NOT(a)))
0t010	$[0,+,0]$	2	NOT(AND(a,a))	3	DIF(a,DIF(a,NOT(a)))	4	SYN(SYN(ANT(a),NOT(a)),a)
0t011	$[0,+,+]$	2	NOT(ANT(a))	3	NOT(DIF(DIF(a,a),a))	2	NOT(ANT(a))
0t012	$[0,+,-]$	3	DIF(NOT(a),AND(a,a))	6	DIF(DIF(NOT(a),a),DIF(NOT(NOT(a)),a))	8	ANT(SYN(NOT(NOT(ANT(a))),SYN(ANT(NOT(a)),a)))
0t020	$[0,-,0]$	3	ANT(NOT(AND(a,a)))	3	DIF(DIF(a,NOT(a)),a)	5	ANT(SYN(SYN(ANT(a),NOT(a)),a))
0t021	$[0,-,+]$	3	DIF(AND(a,a),NOT(a))	6	DIF(DIF(NOT(NOT(a)),a),DIF(NOT(a),a))	7	SYN(NOT(NOT(ANT(a))),SYN(ANT(NOT(a)),a))
0t022	$[0,-,-]$	3	ANT(NOT(ANT(a)))	4	DIF(DIF(DIF(a,a),NOT(a)),a)	3	ANT(NOT(ANT(a)))
0t100	$[+,0,0]$	2	DIF(AND(a,a),a)	3	DIF(NOT(NOT(a)),a)	3	NOT(NOT(ANT(a)))
0t101	$[+,0,+]$	1	AND(a,a)	4	NOT(DIF(a,DIF(a,NOT(a))))	5	NOT(SYN(SYN(ANT(a),NOT(a)),a))
0t102	$[+,0,-]$	1	ANT(a)	2	DIF(DIF(a,a),a)	1	ANT(a)
0t110	$[+,+,0]$	1	NOT(a)	1	NOT(a)	1	NOT(a)
0t111	$[+,+,+]$	2	NOT(DIF(a,a))	2	NOT(DIF(a,a))	3	NOT(ANT(NOT(a)))
0t112	$[+,+,-]$	2	DIF(NOT(a),a)	2	DIF(NOT(a),a)	3	SYN(ANT(a),NOT(a))
0t120	$[+,-,0]$	4	DIF(AND(a,a),NOT(ANT(a)))	7	DIF(DIF(DIF(NOT(NOT(a)),a),DIF(NOT(a),a)),a)	7	SYN(ANT(SYN(NOT(ANT(a)),a)),NOT(NOT(a)))
0t121	$[+,-,+]$	4	DIF(AND(a,a),NOT(AND(a,a)))	8	DIF(NOT(DIF(a,DIF(a,NOT(a)))),DIF(a,DIF(a,NOT(a))))	10	SYN(NOT(NOT(a)),SYN(ANT(SYN(NOT(ANT(a)),a)),NOT(NOT(a))))
0t122	$[+,-,-]$	4	ANT(SYN(NOT(AND(a,a)),a))	4	DIF(DIF(DIF(a,NOT(a)),a),a)	4	ANT(SYN(NOT(ANT(a)),a))
0t200	$[-,0,0]$	2	AND(NOT(a),a)	3	DIF(a,NOT(NOT(a)))	4	ANT(NOT(NOT(ANT(a))))
0t201	$[-,0,+]$	0	a	0	a	0	a
0t202	$[-,0,-]$	2	AND(ANT(a),a)	6	DIF(DIF(a,DIF(a,NOT(a))),NOT(DIF(a,a)))	6	ANT(NOT(SYN(SYN(ANT(a),NOT(a)),a)))
0t210	$[-,+,0]$	4	DIF(NOT(ANT(a)),AND(a,a))	7	DIF(DIF(a,DIF(DIF(a,NOT(a)),a)),NOT(NOT(a)))	7	SYN(ANT(NOT(NOT(a))),SYN(NOT(ANT(a)),a))
0t211	$[-,+,+]$	3	SYN(NOT(AND(a,a)),a)	4	DIF(a,DIF(DIF(a,NOT(a)),a))	3	SYN(NOT(ANT(a)),a)
0t212	$[-,+,-]$	4	DIF(NOT(AND(a,a)),AND(a,a))	8	DIF(DIF(a,DIF(a,NOT(a))),NOT(DIF(a,DIF(a,NOT(a)))))	11	ANT(SYN(NOT(NOT(a)),SYN(ANT(SYN(NOT(ANT(a)),a)),NOT(NOT(a)))))
0t220	$[-,-,0]$	2	ANT(NOT(a))	3	DIF(DIF(a,a),NOT(a))	2	ANT(NOT(a))
0t221	$[-,-,+]$	2	DIF(a,NOT(a))	2	DIF(a,NOT(a))	3	SYN(ANT(NOT(a)),a)
0t222	$[-,-,-]$	3	ANT(NOT(DIF(a,a)))	4	DIF(DIF(a,a),NOT(DIF(a,a)))	4	ANT(NOT(ANT(NOT(a))))

S6. Classification of all 127 nonempty subsets

For each subset: complete without nullary constants (Theorem 6) and its certificate of incompleteness (an invariant shared by all its operations), and complete when $0, +1, -1$ are nullary operations (Remark 3 of the paper) with the reason if not. The criterion of Proposition 6 (the subset contains DIF, or contains SYN and one more operation) reproduces the last two columns on all 127 subsets (checked by `verify_direction_rotation.py`); the proof certificates are: a monotone-operations argument for $\{\mathsf{SYN}\}$ and preservation of ρ (S7) for the subsets of $\{\mathsf{OR}, \mathsf{XOR}, \mathsf{AND}, \mathsf{NOT}, \mathsf{ANT}\}$. Both columns were computed by two methods that agree on every subset (`classification_127.csv`, `classification_table.py`), and by a third script (`independent_free_constants.py`).

#	subset	no const.	certificate	with const.	reason
1	SYN	no	zero-fixing; preserves $\{0,+1\}$	no	unary closure has 10 of 27 functions
2	DIF	no	zero-fixing	yes	
3	OR	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
4	XOR	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
5	AND	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	unary closure has 7 of 27 functions
6	NOT	no	preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
7	ANT	no	zero-fixing; preserves ρ	no	no onto essentially binary operation
8	SYN, DIF	no	zero-fixing	yes	
9	SYN, OR	no	zero-fixing; preserves $\{0,+1\}$	yes	
10	SYN, XOR	no	zero-fixing; preserves $\{0,+1\}$	yes	
11	SYN, AND	no	zero-fixing; preserves $\{0,+1\}$	yes	
12	SYN, NOT	no	preserves $\{0,+1\}$	yes	
13	SYN, ANT	no	zero-fixing	yes	
14	DIF, OR	no	zero-fixing	yes	
15	DIF, XOR	no	zero-fixing	yes	
16	DIF, AND	no	zero-fixing	yes	
17	DIF, NOT	yes		yes	
18	DIF, ANT	no	zero-fixing	yes	
19	OR, XOR	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
20	OR, AND	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
21	OR, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
22	OR, ANT	no	zero-fixing; preserves ρ	no	no onto essentially binary operation
23	XOR, AND	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
24	XOR, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
25	XOR, ANT	no	zero-fixing; preserves ρ	no	no onto essentially binary operation
26	AND, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
27	AND, ANT	no	zero-fixing; preserves ρ	no	unary closure has 7 of 27 functions
28	NOT, ANT	no	preserves ρ	no	no onto essentially binary operation
29	SYN, DIF, OR	no	zero-fixing	yes	
30	SYN, DIF, XOR	no	zero-fixing	yes	
31	SYN, DIF, AND	no	zero-fixing	yes	
32	SYN, DIF, NOT	yes		yes	
33	SYN, DIF, ANT	no	zero-fixing	yes	
34	SYN, OR, XOR	no	zero-fixing; preserves $\{0,+1\}$	yes	
35	SYN, OR, AND	no	zero-fixing; preserves $\{0,+1\}$	yes	
36	SYN, OR, NOT	no	preserves $\{0,+1\}$	yes	
37	SYN, OR, ANT	no	zero-fixing	yes	
38	SYN, XOR, AND	no	zero-fixing; preserves $\{0,+1\}$	yes	
39	SYN, XOR, NOT	no	preserves $\{0,+1\}$	yes	
40	SYN, XOR, ANT	no	zero-fixing	yes	
41	SYN, AND, NOT	no	preserves $\{0,+1\}$	yes	
42	SYN, AND, ANT	no	zero-fixing	yes	
43	SYN, NOT, ANT	yes		yes	
44	DIF, OR, XOR	no	zero-fixing	yes	
45	DIF, OR, AND	no	zero-fixing	yes	
46	DIF, OR, NOT	yes		yes	
47	DIF, OR, ANT	no	zero-fixing	yes	
48	DIF, XOR, AND	no	zero-fixing	yes	
49	DIF, XOR, NOT	yes		yes	
50	DIF, XOR, ANT	no	zero-fixing	yes	
51	DIF, AND, NOT	yes		yes	
52	DIF, AND, ANT	no	zero-fixing	yes	
53	DIF, NOT, ANT	yes		yes	
54	OR, XOR, AND	no	zero-fixing; preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
55	OR, XOR, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	no onto essentially binary operation
56	OR, XOR, ANT	no	zero-fixing; preserves ρ	no	no onto essentially binary operation
57	OR, AND, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
58	OR, AND, ANT	no	zero-fixing; preserves ρ	no	unary closure has 17 of 27 functions
59	OR, NOT, ANT	no	preserves ρ	no	no onto essentially binary operation
60	XOR, AND, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
61	XOR, AND, ANT	no	zero-fixing; preserves ρ	no	unary closure has 17 of 27 functions
62	XOR, NOT, ANT	no	preserves ρ	no	no onto essentially binary operation
63	AND, NOT, ANT	no	preserves ρ	no	unary closure has 17 of 27 functions
64	SYN, DIF, OR, XOR	no	zero-fixing	yes	
65	SYN, DIF, OR, AND	no	zero-fixing	yes	
66	SYN, DIF, OR, NOT	yes		yes	

#	subset	no const.	certificate	with const.	reason
67	SYN, DIF, OR, ANT	no	zero-fixing	yes	
68	SYN, DIF, XOR, AND	no	zero-fixing	yes	
69	SYN, DIF, XOR, NOT	yes		yes	
70	SYN, DIF, XOR, ANT	no	zero-fixing	yes	
71	SYN, DIF, AND, NOT	yes		yes	
72	SYN, DIF, AND, ANT	no	zero-fixing	yes	
73	SYN, DIF, NOT, ANT	yes		yes	
74	SYN, OR, XOR, AND	no	zero-fixing; preserves $\{0,+1\}$	yes	
75	SYN, OR, XOR, NOT	no	preserves $\{0,+1\}$	yes	
76	SYN, OR, XOR, ANT	no	zero-fixing	yes	
77	SYN, OR, AND, NOT	no	preserves $\{0,+1\}$	yes	
78	SYN, OR, AND, ANT	no	zero-fixing	yes	
79	SYN, OR, NOT, ANT	yes		yes	
80	SYN, XOR, AND, NOT	no	preserves $\{0,+1\}$	yes	
81	SYN, XOR, AND, ANT	no	zero-fixing	yes	
82	SYN, XOR, NOT, ANT	yes		yes	
83	SYN, AND, NOT, ANT	yes		yes	
84	DIF, OR, XOR, AND	no	zero-fixing	yes	
85	DIF, OR, XOR, NOT	yes		yes	
86	DIF, OR, XOR, ANT	no	zero-fixing	yes	
87	DIF, OR, AND, NOT	yes		yes	
88	DIF, OR, AND, ANT	no	zero-fixing	yes	
89	DIF, OR, NOT, ANT	yes		yes	
90	DIF, XOR, AND, NOT	yes		yes	
91	DIF, XOR, AND, ANT	no	zero-fixing	yes	
92	DIF, XOR, NOT, ANT	yes		yes	
93	DIF, AND, NOT, ANT	yes		yes	
94	OR, XOR, AND, NOT	no	preserves $\{0,+1\}$; preserves ρ	no	unary closure has 17 of 27 functions
95	OR, XOR, AND, ANT	no	zero-fixing; preserves ρ	no	unary closure has 17 of 27 functions
96	OR, XOR, NOT, ANT	no	preserves ρ	no	no onto essentially binary operation
97	OR, AND, NOT, ANT	no	preserves ρ	no	unary closure has 17 of 27 functions
98	XOR, AND, NOT, ANT	no	preserves ρ	no	unary closure has 17 of 27 functions
99	SYN, DIF, OR, XOR, AND	no	zero-fixing	yes	
100	SYN, DIF, OR, XOR, NOT	yes		yes	
101	SYN, DIF, OR, XOR, ANT	no	zero-fixing	yes	
102	SYN, DIF, OR, AND, NOT	yes		yes	
103	SYN, DIF, OR, AND, ANT	no	zero-fixing	yes	
104	SYN, DIF, OR, NOT, ANT	yes		yes	
105	SYN, DIF, XOR, AND, NOT	yes		yes	
106	SYN, DIF, XOR, AND, ANT	no	zero-fixing	yes	
107	SYN, DIF, XOR, NOT, ANT	yes		yes	
108	SYN, DIF, AND, NOT, ANT	yes		yes	
109	SYN, OR, XOR, AND, NOT	no	preserves $\{0,+1\}$	yes	
110	SYN, OR, XOR, AND, ANT	no	zero-fixing	yes	
111	SYN, OR, XOR, NOT, ANT	yes		yes	
112	SYN, OR, AND, NOT, ANT	yes		yes	
113	SYN, XOR, AND, NOT, ANT	yes		yes	
114	DIF, OR, XOR, AND, NOT	yes		yes	
115	DIF, OR, XOR, AND, ANT	no	zero-fixing	yes	
116	DIF, OR, XOR, NOT, ANT	yes		yes	
117	DIF, OR, AND, NOT, ANT	yes		yes	
118	DIF, XOR, AND, NOT, ANT	yes		yes	
119	OR, XOR, AND, NOT, ANT	no	preserves ρ	no	unary closure has 17 of 27 functions
120	SYN, DIF, OR, XOR, AND, NOT	yes		yes	
121	SYN, DIF, OR, XOR, AND, ANT	no	zero-fixing	yes	
122	SYN, DIF, OR, XOR, NOT, ANT	yes		yes	
123	SYN, DIF, OR, AND, NOT, ANT	yes		yes	
124	SYN, DIF, XOR, AND, NOT, ANT	yes		yes	
125	SYN, OR, XOR, AND, NOT, ANT	yes		yes	
126	DIF, OR, XOR, AND, NOT, ANT	yes		yes	
127	SYN, DIF, OR, XOR, AND, NOT, ANT	yes		yes	

S7. The invariants used for incompleteness

operation	fixes 0	preserves $\{0,+1\}$	preserves ρ
SYN	yes	yes	no
DIF	yes	no	no
OR	yes	yes	yes
XOR	yes	yes	yes
AND	yes	yes	yes
NOT	no	yes	yes
ANT	yes	no	yes

Here $\rho = \{(a,b) \in D^2 : ab \geq 0\}$ has 7 elements and an operation preserves it if it maps every k-tuple of pairs of ρ to a pair of ρ (checked over the 7^2 or 7 cases); the constants $c \in D$ preserve it because $(c,c) \in \rho$. Witnesses: $\mathsf{DIF}(0,+1) = -1 \notin \{0,+1\}$; $(+1,0),(0,-1) \in \rho$ but $(\mathsf{SYN}(+1,0),\mathsf{SYN}(0,-1)) = (+1,-1) \notin \rho$.

S8. The four BID states, involution, difference lift and orbits

The four states of a BID (X,Y) with the difference view $d = X - Y$, the sum view $c = X + Y$ and the total selector of Section VI of the paper (S_c is the one-hot branch selector; $(1,1)$ goes to S_0 by convention):

(X,Y)	d	c	selector
$(0,0)$	0	0	S_0
$(0,1)$	$-$	1	S_{-1}
$(1,0)$	$+$	1	S_{+1}
$(1,1)$	0	2	S_0 (convention)

$\Phi(a,b) = (\mathsf{SYN}(a,b), \mathsf{DIF}(a,b))$ on D^2 (asserted: $\Phi \circ \Phi = \mathrm{id}$):

(a,b)	$\Phi(a,b)$
$(-,-)$	$(-,0)$
$(-,0)$	$(-,-)$
$(-,+)$	$(0,-)$
$(0,-)$	$(-,+)$
$(0,0)$	$(0,0)$
$(0,+)$	$(+,-)$
$(+,-)$	$(0,+)$
$(+,0)$	$(+,+)$
$(+,+)$	$(+,0)$

Difference lift for $n = 1$: the 27 unary functions have $z = 0$: 8 functions, $2^z = 1$ pairs each, $z = 1$: 12 functions, $2^z = 2$ pairs each, $z = 2$: 6 functions, $2^z = 4$ pairs each, $z = 3$: 1 functions, $2^z = 8$ pairs each; the total is $64 = 8^2$ pairs of $\{0,1\}$-valued functions (the 8 XY3 masks). For $n = 2$: 512 masks, 262 144 pairs (all enumerated by `check_c9_chain.py`). Orbits of binary truth tables under the order-8 group generated by SWAP, τ_A, τ_B: 2862 in total, namely 27 of size 1, 54 of size 2, 675 of size 4, 2106 of size 8; cycle counts of the 8 group elements: 3, 3, 5, 6, 6, 6, 6, 9.

S9. A worked example of the three-slice C9 term

For the ternary sum modulo 3, $F(a,b) = ((a+b+1) \bmod 3) - 1$, the three branches $G_c = F(c,\cdot)$ are unary functions with the minimal terms below (variable b), and $P_c = \mathsf{AND}(E_c(a), G_c(b))$ with $E_c(x) = \mathsf{NOT}(\mathsf{XOR}(x,c))$, where the constants are the terms $0 = \mathsf{DIF}(x,x)$, $+1 = \mathsf{NOT}(0)$, $-1 = \mathsf{ANT}(+1)$:

c	G_c	term of G_c	P_c as a term
$-$	$[+,-,0]$	`DIF(AND(b,b),NOT(ANT(b)))`	`AND(NOT(XOR(a,ANT(NOT(DIF(a,a))))),DIF(AND(b,b),NOT(ANT(b))))`
0	$[-,0,+]$	`b`	`AND(NOT(XOR(a,DIF(a,a))),b)`
$+$	$[0,+,-]$	`DIF(NOT(b),AND(b,b))`	`AND(NOT(XOR(a,NOT(DIF(a,a)))),DIF(NOT(b),AND(b,b)))`

Then $F(a,b) = \mathsf{SYN}\big(P_{-1}, \mathsf{SYN}(P_0, P_{+1})\big)$. The generator evaluates this expression at all nine points and asserts that it equals F. `check_c9_chain.py` does the same for all 19 683 binary functions.

S10. Nine-trit configuration words (T9ID)

Row-major reading of the truth table (rows $a = -,0,+$, columns $b = -,0,+$), digits $0,1,2$ for $0,+1,-1$. The word of SYN is the word that the published divisional CN 122173050 A lists for its operation named OR; the words of DIF and AND agree with it. Words of the first six operations marked below match `results.json`.

operation	T9ID word	rows $a=-,0,+$ (columns $b=-,0,+$)
SYN	`0t220_201_011`	$--0\ -0+\ 0++$
DIF	`0t022_102_110`	$0--\ +0-\ ++0$
OR	`0t110_101_011`	$++0\ +0+\ 0++$
XOR	`0t011_101_110`	$0++\ +0+\ ++0$
AND	`0t102_000_201`	$+0-\ 000\ -0+$
MAX	`0t201_001_111`	$-0+\ 00+\ +++$
MIN	`0t222_200_201`	$---\ -00\ -0+$
SUM (mod 3)	`0t120_201_012`	$+-0\ -0+\ 0+-$
CARRY	`0t200_000_001`	$-00\ 000\ 00+$

S11. Where each claim of the paper is checked

claim	script	output key
Lemma 1, Theorem 1 (parity, polarization)	`chai_checker.py; check_c9_chain.py; chai_verify_v1_increments.py`	identities.parity, identities.polarization; v1: parity and polarization at all nine pairs
Theorem 2 (involution)	`chai_checker.py; chai_verify_v1_increments.py`	identities.involution_fixed_points
Proposition 1 (minimal carrier), the three sections	`chai_verify_v1_increments.py`	chaiv1_increments_results.json (closure iteration, section vectors, involution test)
Remark 2 (table symmetry), Proposition 5 (fold), S14–S16	`verify_direction_rotation.py; chai_verify_v1_increments.py`	direction_rotation_results.json (115 tests, negative controls included); v1: clockwise rotation of the five tables, fold congruence
Lemma 2, Theorems 3 and 4 (explicit terms)	`chai_checker.py; chai_verify_v1_increments.py`	paper_terms; DIF_NOT_complete_constructive; Z_n_from_five_operations; DIF_odd; v1: all 6561 functions of Z_2, all 19683 binary functions (Thm 3c), all 81 odd functions from DIF alone
Theorem 5 (Boolean layer), Table III	`chai_verify_v1_increments.py`	v1: all 3888 functions of B_2 and 300 random functions of B_3 rebuilt; class sizes and the printed numbers of Table III
Table II (layers)	`chai_verify_v1_increments.py`	v1: parity, range, zero-fixing, B-keeping and rho-keeping computed for the seven primitives and compared with the printed rows
Theorem 3(b): $\mathscr{Z}_n$ from five operations	`chai_checker.py --full`	closure SYN+DIF+OR+XOR+AND (6561 functions)
Theorem 6: 40 of 127	`chai_checker.py; classification_table.py; independent_free_constants.py; chai_verify_v1_increments.py`	classification; classification_127.csv (per-subset, two methods plus an independent script); v1: the criterion agrees with the csv
Remark 3: 95 with constants	`chai_checker.py; results_consts_*.json; verify_direction_rotation.py`	free_constants; closure with free constants *; direction_rotation_results.json (terms, invariants, 127-subset criterion)
Remark 1 (sign rule of AND)	`verify_direction_rotation.py; chai_verify_v1_increments.py`	direction_rotation_results.json; failure at (2,1) on -2..2
Connectives as terms (S3), Proposition 3 (K_3)	`chai_checker.py; chai_verify_v1_increments.py; this supplement`	paper_terms; shortest_terms; every term evaluated by this generator; v1: MAX, MIN terms
27 unary terms (S5)	`chai_checker.py; independent_check.py`	unary_catalogue.json; catalogue mismatches 0
Difference lift (Prop. 6)	`chai_checker.py; check_c9_chain.py; chai_verify_v1_increments.py`	difference_lift; n=2 pairs enumerated
2862 orbits (Prop. 7)	`chai_checker.py; verify_direction_rotation.py; chai_verify_v1_increments.py`	orbits; direction_rotation_results.json; v1: group of order 8, Burnside sum
T9ID words	`chai_checker.py; chai_verify_v1_increments.py; this supplement`	T9ID_words
Table IV (closures with the constants), Remark 3	`chai_verify_v1_increments.py; chai_checker.py --full-consts`	results_consts_*.json (size and heights of the five closures); v1: the printed table against these files
XY3 and G3 tables, hexaogram and bi-trigram symbols (S17)	`verify_direction_rotation.py; this supplement`	direction_rotation_results.json (S17 tests); every row asserted by the generator
C9 three-slice selector (19683 functions)	`check_c9_chain.py; chai_verify_v1_increments.py`	all binary functions rebuilt as $U-V$ and as the term of Section VI; v1: Shannon/Post expansion on 500 random functions

S12. What the computations establish, and what they do not

Exhaustive: the parity lemma and polarization on all 9 pairs; the involution on all 9 points; every function of $\mathscr{Z}_2$ (6561) rebuilt from the five operations; all 19683 binary and 27 unary functions rebuilt from $\{\mathsf{DIF},\mathsf{NOT}\}$; the closures of S4; the classification of all 127 subsets by two methods; the difference lift for $n=1$ and $n=2$ (all 64 and 262 144 pairs); all 19683 binary functions as three-slice C9 terms. **Random samples (fixed seed 7):** 25 functions each of arity 3 and 4 rebuilt from $\{\mathsf{DIF},\mathsf{NOT}\}$; 25 functions each of $\mathscr{Z}_3$ and $\mathscr{Z}_4$ from the five operations; 25 odd functions each of arity 1, 3, 4 from DIF. These are evidence, not proofs. **Direction, rotation and folding (S14–S16):** the program `verify_direction_rotation.py` (standard library only, 115 tests, Python 3.9.6) checks the table cycle, the lemma

$\mathscr{R}(T_F) = T_{F\circ\rho}$ for all 19683 functions, $\mathscr{R}Z = -iZ$ in exact Gaussian integers, the folding statements, the congruences, subalgebras and automorphisms by exhaustion, the Boolean and set identities, the C_4 and dihedral orbit counts, and the 17 sign-linear functions; it includes negative controls (wrong sign, wrong direction, reflection for rotation) that must come out false. **Proofs:** the statements for every arity n (Theorems 3, 4 and 5) rest on the explicit constructions of Lemma 2 and Theorems 3, 4 and 5 in the paper, not on the sampled cases; the code checks the building blocks (S3) exhaustively, and S18 evaluates the constructions on every binary function of the classes. Environment: Python 3.9.6, numpy 2.0.2; the full run took about 82 seconds on one core. Hashes of all files of the package are in `SHA256SUMS`.

S13. Use of generative AI tools

Generative AI tools, including ChatGPT by OpenAI, Claude by Anthropic, and Gemini by Google, were used to assist with language editing, to discuss candidate proof formulations, and to develop and review the verification scripts described in this supplement. The author independently checked the mathematical arguments, the cited sources and the computational results, and accepts responsibility for the final content.

S14. The quarter turn of a truth table

Convention. The matrix T_F of a binary F has rows $a = -, 0, +$ from top to bottom and columns $b = -, 0, +$ from left to right. The clockwise quarter turn is

$$\mathscr{R}\begin{pmatrix} p & q & r \\ s & t & u \\ v & w & x \end{pmatrix} = \begin{pmatrix} v & s & p \\ w & t & q \\ x & u & r \end{pmatrix},$$

$\mathscr{R}^{-1}$ is the counterclockwise turn, and $-T$ negates every entry. Put $\rho(a,b) = (-b,a)$ (the quarter turn of the argument grid; $a+ib \mapsto i(a+ib)$).

Proposition S14 (quarter turn). For every $F : D^2 \to D$, $\mathscr{R}(T_F) = T_{F\circ\rho}$. Hence $\mathscr{R}T_{\mathsf{SYN}} = T_{\mathsf{DIF}}$, $\mathscr{R}T_{\mathsf{DIF}} = -T_{\mathsf{SYN}}$, $\mathscr{R}T_{\mathsf{OR}} = T_{\mathsf{XOR}}$, $\mathscr{R}T_{\mathsf{XOR}} = T_{\mathsf{OR}}$, $\mathscr{R}T_{\mathsf{AND}} = -T_{\mathsf{AND}}$, $\mathscr{R}^4 = \mathrm{id}$, and $\mathscr{R}^2 T_F = -T_F$ iff F is odd ($\mathscr{R}^2 T_F = T_F$ iff F is even).

Proof. The entry of $\mathscr{R}(M)$ in row a and column b is the entry of M in the row labelled $-b$ and the column labelled a; for $M = T_F$ this is $F(-b,a)$. Then $\mathsf{SYN}(-b,a) = \mathrm{sgn}(a-b)$, $\mathsf{DIF}(-b,a) = \mathrm{sgn}(-b-a)$, $\mathsf{OR}(-b,a) = \mathrm{sgn}((a-b)^2)$, $\mathsf{XOR}(-b,a) = \mathrm{sgn}((a+b)^2)$, $\mathsf{AND}(-b,a) = \mathrm{sgn}(-4ab)$. Since $\rho^2(a,b) = (-a,-b)$, $\mathscr{R}^2 T_F = T_{F(-a,-b)}$. □

The generator rotates the matrices below and asserts every equality; the program `verify_direction_rotation.py` checks the lemma $\mathscr{R}(T_F) = T_{F\circ\rho}$ for all 19683 binary F.

T_{SYN}

	−	0	+
−	−	−	0
0	−	0	+
+	0	+	+

→ $T_{\mathsf{DIF}} = \mathscr{R}T_{\mathsf{SYN}}$

	−	0	+
−	0	−	−
0	+	0	−
+	+	+	0

→ $-T_{\mathsf{SYN}} = \mathscr{R}T_{\mathsf{DIF}}$

	−	0	+
−	+	+	0
0	+	0	−
+	0	−	−

→ $-T_{\mathsf{DIF}}$

	−	0	+
−	0	+	+
0	−	0	+
+	−	−	0

T_{OR}

	−	0	+
−	+	+	0
0	+	0	+
+	0	+	+

$T_{\mathsf{XOR}} = \mathscr{R}T_{\mathsf{OR}}$

	−	0	+
−	0	+	+
0	+	0	+
+	+	+	0

T_{AND}

	−	0	+
−	+	0	−
0	0	0	0
+	−	0	+

$\mathscr{R}T_{\mathsf{AND}} = -T_{\mathsf{AND}}$

	−	0	+
−	−	0	+
0	0	0	0
+	+	0	−

The identities of Proposition S14 for the sgn-defined operations hold on every symmetric grid $\{-m,\dots,m\}$ (checked for $m = 2,3,5$; they follow from the same one-line computation), whereas the polarization identity of Theorem 1 does not (Remark 1): only the descent to signs is special to D. The even operations $\mathsf{OR}, \mathsf{XOR}$ have period 2 under $\mathscr{R}$ and the odd operations $\mathsf{SYN}, \mathsf{DIF}$ period 4, because $\mathscr{R}^2$ is the substitution $(a,b) \mapsto (-a,-b)$. Among the 19683 binary functions exactly $81 = 3^4$ are odd and $243 = 3^5$ are even.

Three notions to keep apart. (1) turning the picture of the table, (2) substituting ρ in the arguments (equal to (1) by Proposition S14), (3) identities such as $\mathsf{DIF}(a,b) = \mathsf{SYN}(a,\mathsf{ANT}(b))$, which negate one argument and *reflect* the table (reverse the columns). For commutative F the substitution $(-b,a)$ and $(a,-b)$ coincide, which is why rotation and reflection agree on SYN and OR; DIF is antisymmetric, so rotation gives $-\mathsf{SYN}$ and reflection gives $+\mathsf{SYN}$:

$\mathsf{DIF}(-b,a)$: rotation

	−	0	+
−	+	+	0
0	+	0	−
+	0	−	−

$\mathsf{DIF}(a,-b)$: reflection

	−	0	+
−	−	−	0
0	−	0	+
+	0	+	+

$\mathsf{SYN}(-b,a)$: rotation

	−	0	+
−	0	−	−
0	+	0	−
+	+	+	0

$\mathsf{SYN}(a,-b)$: reflection

	−	0	+
−	0	−	−
0	+	0	−
+	+	+	0

Orbits. Under the rotation group $C_4 = \langle \rho \rangle$ the 19683 binary functions fall into $4995 = \frac{1}{4}(3^9 + 2\cdot 3^3 + 3^5)$ orbits (27 of size 1, 108 of size 2, 4860 of size 4); under the dihedral group $\langle \rho, \mathsf{SWAP} \rangle$ of order 8 (the group of S8) into 2862 orbits (27 of size 1, 54 of size 2, 675 of size 4, 2106 of size 8), as in Section VI of the paper. The orbit of DIF is $\{\mathsf{DIF}, \mathsf{SYN}, -\mathsf{DIF}, -\mathsf{SYN}\}$, that of OR is $\{\mathsf{OR}, \mathsf{XOR}\}$ and that of AND is $\{\mathsf{AND}, -\mathsf{AND}\}$ (all enumerated).

Sign-linear binary functions (degree one). The functions $\operatorname{sgn}(\alpha a + \beta b)$, $\alpha, \beta \in \mathbb{Z}$, are exactly 17: the value at the eight non-zero points of the grid is $\pm\operatorname{sgn}\alpha$, $\pm\operatorname{sgn}\beta$, $\pm\operatorname{sgn}(\alpha+\beta)$, $\pm\operatorname{sgn}(\alpha-\beta)$, so a function is determined by the signs of these four numbers, and the directions of (α, β) form 8 open sectors, 8 rays and the origin. They form dihedral orbits of sizes 8 (sectors), 4 ($\mathsf{SYN}, \mathsf{DIF}, -\mathsf{SYN}, -\mathsf{DIF}$), 4 ($\pm a, \pm b$) and 1 (zero). All are odd, hence lie in the clone generated by DIF (Theorem 4). The program enumerates $|\alpha|, |\beta| \le 12$ and finds 17 functions; the count for all integers follows from the sign argument. This is an elementary observation, not a result used in the paper.

(α, β)	T9ID word	rows $a = -, 0, +$
$(0,0)$	`0t000_000_000`	000 000 000
$(0,1)$	`0t201_201_201`	−0+ −0+ −0+
$(1,0)$	`0t222_000_111`	− − − 000 + + +
$(1,-1)$	`0t022_102_110`	0 − − +0− + +0
$(1,1)$	`0t220_201_011`	− −0 −0+ 0+ +
$(1,2)$	`0t221_201_211`	− − + −0+ − + +
$(2,-1)$	`0t222_102_111`	− − − +0− + + +
$(2,1)$	`0t222_201_111`	− − − −0+ + + +
$(-12,0)$	`0t111_000_222`	+ + + 000 − − −
$(0,-12)$	`0t102_102_102`	+0− +0− +0−
$(-12,1)$	`0t111_201_222`	+ + + −0+ − − −
$(1,-12)$	`0t122_102_112`	+ − − +0− + + −
$(-12,-11)$	`0t111_102_222`	+ + + +0− − − −
$(-11,-12)$	`0t112_102_122`	+ + − +0− + − −
$(-11,12)$	`0t211_201_221`	− + + −0+ − − +
$(-12,-12)$	`0t110_102_022`	+ +0 +0− 0 − −
$(-12,12)$	`0t011_201_220`	0 + + −0+ − −0

S15. The complex table $Z = T_{\mathsf{SYN}} + i\,T_{\mathsf{DIF}}$

Let $\operatorname{sgn}_c(x+iy) = \operatorname{sgn}x + i\operatorname{sgn}y$ on Gaussian integers, and let $\mathscr{R}$ act on a complex matrix by acting on its real and imaginary parts (the linear extension).

Statement (paper, Section III-C). $\mathscr{R}^{-1}Z = iZ$ and $\mathscr{R}Z = -iZ$; hence $\mathscr{R}^4 Z = Z$ and $\mathscr{R}^2 Z = -Z$.

Proof. (a) $\operatorname{sgn}_c(iu) = i\operatorname{sgn}_c(u)$: multiplication by i sends (x,y) to $(-y,x)$ and sgn is odd; applying it to $-iu$ gives $\operatorname{sgn}_c(-iu) = -i\operatorname{sgn}_c(u)$. (b) $Z(a,b) = \operatorname{sgn}_c\big((a+b)+i(a-b)\big) = \operatorname{sgn}_c\big((1+i)\overline{(a+ib)}\big)$, because $(1+i)(a-ib) = (a+b)+i(a-b)$. (c) With $w = a+ib$, ρ is $w \mapsto iw$ and $\overline{iw} = -i\bar{w}$, so $Z(\rho(a,b)) = \operatorname{sgn}_c\big(-i(1+i)\bar{w}\big) = -iZ(a,b)$. By Proposition S14 of the paper applied to the real and imaginary parts, $\mathscr{R}Z = Z\circ\rho = -iZ$, and $\mathscr{R}^{-1}Z = iZ$ follows. □

The program checks (a) for all Gaussian integers with $|x|, |y| \le 6$, (b) at the nine points, (c) for all w with $|a|, |b| \le 6$ (the formula of (b) extended to all integers), and the proposition entry by entry in exact integer arithmetic; wrong-sign versions ($\mathscr{R}Z = iZ$, $\mathscr{R}^{-1}Z = -iZ$) are checked to be *false*.

	$Z = T_{\mathsf{SYN}} + iT_{\mathsf{DIF}}$			$iZ = \mathscr{R}^{-1}Z$		
	$-$	0	$+$	$-$	0	$+$
$-$	-1	$-1-i$	$-i$	$-i$	$1-i$	1
0	$-1+i$	0	$1-i$	$-1-i$	0	$1+i$
$+$	i	$1+i$	1	-1	$-1+i$	i

BID view. At $(a,b) = (X,Y) \in \{0,1\}^2$ the unsaturated values are $c+id = (1+i)\overline{(X+iY)}$ with $c = X+Y$, $d = X-Y$ (the sum and difference views of a BID), and $Z(X,Y) = \operatorname{sgn}c + i\operatorname{sgn}d$ is their saturation. The identity $c^2 - d^2 = 4XY$ is the BID form of the polarization $\mathsf{AND} = \mathsf{DIF}(\mathsf{OR}, \mathsf{XOR})$.

(X,Y)	c	d	$c+id$
$(0,0)$	0	0	0
$(0,1)$	1	-1	$1-i$
$(1,0)$	1	1	$1+i$
$(1,1)$	2	0	2

After folding. The folded table $W = T_{\mathsf{OR}} + i\,T_{\mathsf{XOR}}$ satisfies $\mathscr{R}W = \mathscr{R}^{-1}W = i\overline{W}$: the turn acts as a reflection (order 2) and not as multiplication by i.

$W = T_{\mathsf{OR}} + iT_{\mathsf{XOR}}$				$i\overline{W} = \mathscr{R}W$			
	$-$	0	$+$		$-$	0	$+$
$-$	1	$1+i$	i	$-$	i	$1+i$	1
0	$1+i$	0	$1+i$	0	$1+i$	0	$1+i$
$+$	i	$1+i$	1	$+$	1	$1+i$	i

What this does and does not say. It says that the two real tables are the two components of one complex-valued table and that the quarter turn of the table acts on it by multiplication by $-i$ (clockwise) or i (counterclockwise), because sgn commutes with the dihedral symmetry of $\mathbb{Z}^2$. It does not make i the cause of the algebra, and it does not give the algebra an interpretation by complex numbers.

S16. Folding, congruences, Boolean restriction and sets

Proposition S16(ii): the kernel of MAG**.** Let $\theta = \{\{-1,+1\},\{0\}\}$ (an equivalence relation, not a congruence of the algebra) be the kernel of $\mathsf{MAG}(a) = a^2$. An operation respects θ if equivalent arguments give equivalent values. Checked over all pairs of θ-equivalent argument tuples:

operation	respects θ	first violation
SYN	no	$\mathsf{MAG}(\mathsf{SYN}(-,-)) = 1 \neq 0 = \mathsf{MAG}(\mathsf{SYN}(-,+))$ although MAG agrees on the arguments
DIF	no	$\mathsf{MAG}(\mathsf{DIF}(-,-)) = 0 \neq 1 = \mathsf{MAG}(\mathsf{DIF}(-,+))$ although MAG agrees on the arguments
OR	no	$\mathsf{MAG}(\mathsf{OR}(-,-)) = 1 \neq 0 = \mathsf{MAG}(\mathsf{OR}(-,+))$ although MAG agrees on the arguments
XOR	no	$\mathsf{MAG}(\mathsf{XOR}(-,-)) = 0 \neq 1 = \mathsf{MAG}(\mathsf{XOR}(-,+))$ although MAG agrees on the arguments
AND	yes	
NOT	no	$\mathsf{MAG}(\mathsf{NOT}(-)) = 1 \neq 0 = \mathsf{MAG}(\mathsf{NOT}(+))$ although MAG agrees on the arguments
ANT	yes	

So MAG is a homomorphism from the reduct $(D;\mathsf{AND},\mathsf{ANT})$ onto $(\{0,1\};\min,\mathrm{id})$, and from no reduct containing another of the seven operations ($\mathsf{MAG}(\mathsf{AND}(a,b)) = \min(\mathsf{MAG}\,a, \mathsf{MAG}\,b)$, $\mathsf{MAG}(\mathsf{ANT}\,a) = \mathsf{MAG}\,a$).

Proposition S16(iii): no nontrivial congruence, proper nonempty subalgebra or nontrivial automorphism. Constants $0,+1,-1$ and all 27 unary functions are terms (Theorem 6 and its proof; S5). *Congruences:* let θ' identify $x \neq y$. A unary term u with $u(x) = 0$, $u(y) = +1$ gives $0\ \theta'\ +1$; for any p,q a unary term v with $v(0) = p$, $v(+1) = q$ gives $p\ \theta'\ q$. *Subalgebras:* a subalgebra containing s contains $c(s)$ for every constant unary term c, hence all of D. *Automorphisms:* an automorphism φ satisfies $\varphi(c(s)) = c(\varphi(s))$ for the constant unary terms, hence fixes every element. The program confirms by exhaustion: of the five partitions of D only the two trivial ones are congruences, no nonempty proper subset is closed, and only the identity of the six permutations is an automorphism. (The same facts hold for any primal algebra; we prove them directly.)

The reducts behave differently: $\{0,+1\}$ is a subalgebra of $(D;\mathsf{SYN},\mathsf{OR},\mathsf{XOR},\mathsf{AND},\mathsf{NOT})$, and not of the full algebra since $\mathsf{DIF}(0,+1) = -1$ and $\mathsf{ANT}(+1) = -1$.

Boolean restriction. On $\{0,1\}$, SYN = OR = Boolean or, XOR = Boolean exclusive or, AND is Boolean and, $\mathsf{NOT}(x) = 1 - x$; DIF leaves the carrier. Two-valued inputs thus produce a signed result outside the carrier, and $\mathsf{MAG} \circ \mathsf{DIF} = \mathsf{XOR}$ folds the two directions into one value. This does not mean that Boolean logic contains -1.

(x,y)	SYN	OR	XOR	AND	NOT(x)	DIF	MAG(DIF)
$(0,0)$	0	0	0	0	1	0	0
$(0,1)$	1	1	1	0	1	$-$	$+$
$(1,0)$	1	1	1	0	0	$+$	$+$
$(1,1)$	1	1	0	1	0	0	0

Sets. For subsets A,B of a universe, the characteristic functions satisfy $\mathbf{1}_{A\cup B} = \mathbf{1}_A \vee \mathbf{1}_B$, $\mathbf{1}_{A\triangle B} = \mathbf{1}_A \oplus \mathbf{1}_B$, $\mathbf{1}_{A\cap B} = \mathbf{1}_A \wedge \mathbf{1}_B$ and $\mathbf{1}_{A\cap B} = \mathbf{1}_{A\cup B} - \mathbf{1}_{A\triangle B}$ (inclusion–exclusion); with OR = SYN, XOR, AND on $\{0,1\}$ this is the polarization identity of Theorem 1, which extends it to signed values. Example ($U = \{1,2,3\}$, $A = \{1,2\}$, $B = \{2,3\}$; columns are the points $1,2,3$; all 64 pairs of subsets of a 3-set are checked by the program):

	1	2	3
A	1	1	0
B	0	1	1
$\mathbf{1}_{A\cup B}$ = OR	1	1	1
$\mathbf{1}_{A\triangle B}$ = XOR	1	0	1
$\mathbf{1}_{A\cap B}$ = AND	0	1	0
DIF(OR,XOR)	0	1	0

The areas of regions in a Venn diagram are not defined here and are not identified with values of the algebra.

Sign rule of AND (Remark 1 of the paper). For all integers, $4ab = (a+b)^2 - (a-b)^2$, hence $\operatorname{sgn}(ab) = \operatorname{sgn}(|a+b| - |a-b|)$: the signs of a product are read off from whether the sum or the difference is the larger in size (equal signs make $|a+b|$ larger). This is the classical quarter-square identity. On D the sizes are $0,1,2$ and by Lemma 1 two different nonzero squares never occur, so the squares can be folded to signs first, which gives $\mathsf{AND} = \mathsf{DIF}(\mathsf{OR},\mathsf{XOR})$ (Theorem 1). On $\{-2,\dots,2\}$ the folded term fails (first failure $(a,b) = (2,1)$: $p^2 - q^2 = 8 > 0$, but $\mathsf{DIF}(1,1) = 0$), while the size comparison itself still holds. The program checks both statements. So the sign rules $-\cdot- = +$, $+\cdot- = -$ are consequences of the tables of SYN, DIF and MAG on D, but the statement "multiplication is nothing but folded sums and differences" is not claimed for other alphabets, and the wording about "aligned axes" is an interpretation.

Status of the readings.

statement	status
$\mathscr{R}(T_F) = T_{F\circ\rho}$, the table cycle, $\mathscr{R}Z = -iZ$	proved (Prop. S14 and S15); finite instances checked by program.
$\mathsf{OR} = \mathsf{MAG}\circ\mathsf{SYN}$, $\mathsf{XOR} = \mathsf{MAG}\circ\mathsf{DIF}$; $\ker\mathsf{MAG}$ respected only by AND, ANT; no nontrivial congruence	proved by evaluation at nine points and a short argument (Prop. S16); checked by program.
Boolean restriction to $\{0,1\}$; set identities $\operatorname{sgn}(ab) = \operatorname{sgn}(\lvert a+b\rvert - \lvert a-b\rvert)$; $\mathsf{AND} = \mathsf{DIF}(\mathsf{OR},\mathsf{XOR})$ on D (Remark 1, Thm. 1)	classical facts expressed in this signature; checked by program. proved; checked by program.
Coin model: $-1,+1$ two faces, 0 the edge; ANT turns the coin, MAG forgets the face; "multiplication emerges from aligned axes"	interpretation, not a theorem. An analogy; a zero need not mean an edge in other settings.
The complex unit i as the origin of the algebra; the algebra as a foundation of geometry, number theory, vectors or mass	*not claimed.* Would need a defined space, distance, area and a proof; a research programme at most.
Sign-degree filtration, equational axiomatization, lattice of the clones generated by the 127 subsets	open.

S17. The XY3 and G3 tables of the published divisional

The eight XY3 masks are labeled by the eight trigrams (top line = bit 2, bottom line = bit 0, solid line = 1), following the labeling of the author's filings; each of the 27 G3 values is drawn below as a hexaogram and as a bi-trigram. The published divisional CN 122173050 A describes three configuration layers, XY3 (eight functions), G3 (27 functions) and C9 (a selector of three G3 branches). This section reproduces the first two from their definitions and places them in the algebra; nothing in the paper depends on it, and the notation x,y is the pair of bits (X,Y) of a BID.

Definitions used. The three BID states are $(X,Y) = (1,0),(0,0),(0,1)$, i.e. $d = X - Y = +1,0,-1$ (the state $(1,1)$ is sent to $d = 0$ by the convention of S8). An XY3 function is a three-bit mask, bit 0 for $(1,0)$, bit 1 for $(0,0)$ and bit 2 for $(0,1)$, read as a Boolean function of x,y on these three states. A G3 function is the difference $\mathrm{XY3}(A) - \mathrm{XY3}(a)$ of two masks A,a with no common one ($A \wedge a = 0$); there are exactly $3^3 = 27$ such pairs, and they give the 27 distinct unary functions of D. The identifier gid $\in \{-13,\dots,13\}$ is the balanced-ternary number whose three digits (most significant first; $0,1,2$ standing for $0,+1,-1$) are the values $g(-1), g(0), g(+1)$ at the states $d = -1,0,+1$. Hence the T3 code of gid is exactly the code of S5, and every G3 branch is a Chai term (last column, the minimal term over all seven operations of S5, variable $a = d$). The generator checks that every row below is computed from these definitions, that the 27 functions are distinct, and that every term evaluates to its function at all three points; `verify_direction_rotation.py` checks the same statements independently.

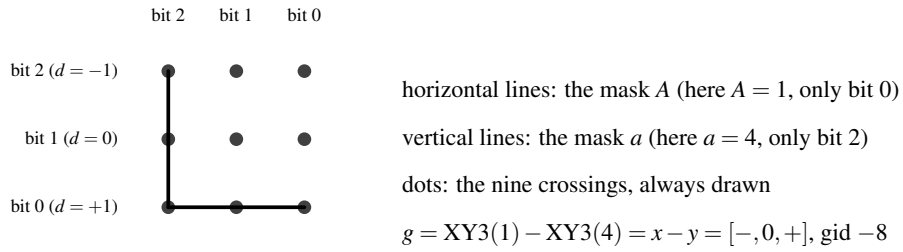


Figure S17. The hexaogram (hexa-o-gram, "o" for orthogonal) of a G3 function, here $(A,a)=(1,4)$: the bits of A are horizontal lines, the bits of a vertical lines.

XY3. Values are those of the mask at the states $(0,1),(0,0),(1,0)$ ($d=-1,0,+1$). The Boolean formulas $x,y,\ldots$ agree with the masks on the three states only; at $(1,1)$ they take other values (for example $x\vee y=1$ there, whereas the zon state has the value 0), which is why the domain is restricted to the three states or $(1,1)$ is sent to the zon state.

id	mask	trigram	function	$d=-1$	$d=0$	$d=+1$
0	`0b000`	☷ Kun	0	0	0	0
1	`0b001`	☳ Zhen	x	0	0	$+$
2	`0b010`	☵ Kan	$x'y'$	0	$+$	0
3	`0b011`	☱ Dui	y'	0	$+$	$+$
4	`0b100`	☶ Gen	y	$+$	0	0
5	`0b101`	☲ Li	$x\vee y$	$+$	0	$+$
6	`0b110`	☴ Xun	x'	$+$	$+$	0
7	`0b111`	☰ Qian	1	$+$	$+$	$+$

G3. Pairs (A,a) are the XY3 ids of the two masks; $g=[g(-1),g(0),g(+1)]$. The *hexaogram* column draws the two masks superposed, as a 3×3 grid of three horizontal and three vertical lines, six lines in all (Figure S17): the three bits of A are the horizontal lines, the three bits of a the vertical lines, and a line is present exactly where its bit is 1; the nine crossing points are always marked by small dots, so that the position of an absent line can be read. The *bi-trigram* column writes the trigram of A and the trigram of a side by side (top line = bit 2). Both symbols are generated from the pair (A,a) by the same program that builds the table, so symbol, code and function cannot disagree.

gid	T3 code	(A,a)	Hexaogram (A,a)	Bi-trigram (A,a)	$\mathrm{XY3}(A)-\mathrm{XY3}(a)$	g	size	Chai term
-13	`0t222`	(0,7)	⫼	☷☰	-1	$[-,-,-]$	3	`ANT(NOT(DIF(a,a)))`
-12	`0t220`	(0,6)	⫻	☷☱	$-x'$	$[-,-,0]$	2	`ANT(NOT(a))`
-11	`0t221`	(1,6)	𝌆	☳☱	$x-x'$	$[-,-,+]$	2	`DIF(a,NOT(a))`
-10	`0t202`	(0,5)	☰	☷☲	$-(x\vee y)$	$[-,0,-]$	2	`AND(ANT(a),a)`
-9	`0t200`	(0,4)	䷀	☷☷	$-y$	$[-,0,0]$	2	`AND(NOT(a),a)`
-8	`0t201`	(1,4)	䷊	☳☷	$x-y$	$[-,0,+]$	0	`a`
-7	`0t212`	(2,5)	䷿	☵☲	$x'y'-(x\vee y)$	$[-,+,-]$	4	`DIF(NOT(AND(a,a)),AND(a,a))`
-6	`0t210`	(2,4)	䷗	☵☷	$x'y'-y$	$[-,+,0]$	4	`DIF(NOT(ANT(a)),AND(a,a))`
-5	`0t211`	(3,4)	䷭	☱☷	$y'-y$	$[-,+,+]$	3	`SYN(NOT(AND(a,a)),a)`
-4	`0t022`	(0,3)	䷫	☷☰	$-y'$	$[0,-,-]$	3	`ANT(NOT(ANT(a)))`
-3	`0t020`	(0,2)	䷆	☷☵	$-x'y'$	$[0,-,0]$	3	`ANT(NOT(AND(a,a)))`
-2	`0t021`	(1,2)	䷮	☳☵	$x-x'y'$	$[0,-,+]$	3	`DIF(AND(a,a),NOT(a))`
-1	`0t002`	(0,1)	䷖	☷☳	$-x$	$[0,0,-]$	3	`ANT(NOT(NOT(a)))`
0	`0t000`	(0,0)	䷁	☷☷	0	$[0,0,0]$	1	`DIF(a,a)`
1	`0t001`	(1,0)	䷒	☳☷	x	$[0,0,+]$	2	`NOT(NOT(a))`
2	`0t012`	(2,1)	䷾	☵☳	$x'y'-x$	$[0,+,-]$	3	`DIF(NOT(a),AND(a,a))`
3	`0t010`	(2,0)	䷜	☵☷	$x'y'$	$[0,+,0]$	2	`NOT(AND(a,a))`
4	`0t011`	(3,0)	䷛	☱☷	y'	$[0,+,+]$	2	`NOT(ANT(a))`
5	`0t122`	(4,3)	䷩	☶☳	$y-y'$	$[+,-,-]$	4	`ANT(SYN(NOT(AND(a,a)),a))`
6	`0t120`	(4,2)	䷦	☶☵	$y-x'y'$	$[+,-,0]$	4	`DIF(AND(a,a),NOT(ANT(a)))`
7	`0t121`	(5,2)	䷷	☲☵	$x\vee y-x'y'$	$[+,-,+]$	4	`DIF(AND(a,a),NOT(AND(a,a)))`
8	`0t102`	(4,1)	䷇	☶☳	$y-x$	$[+,0,-]$	1	`ANT(a)`
9	`0t100`	(4,0)	䷏	☶☷	y	$[+,0,0]$	2	`DIF(AND(a,a),a)`

gid	T3 code	(A,a)	Hexaogram (A,a)	Bi-trigram (A,a)	$\mathrm{XY3}(A)-\mathrm{XY3}(a)$	g	size	Chai term
10	0t101	(5,0)			$x \vee y$	$[+,0,+]$	1	AND(a,a)
11	0t112	(6,1)			$x'-x$	$[+,+,-]$	2	DIF(NOT(a),a)
12	0t110	(6,0)			x'	$[+,+,0]$	1	NOT(a)
13	0t111	(7,0)			1	$[+,+,+]$	2	NOT(DIF(a,a))

Selectors. On the three states the functions x, $x'y'$, y (the masks 1, 2, 4) are one-hot. At $(1,1)$ both x and y equal 1 and $x'y' = 0$, so a sum of the three products with these Boolean selectors would add two branches there; the total selector of the paper (S8) sends $(1,1)$ to the zon branch instead. The C9 layer is the three-branch selection of S9.

S18. Checks added for this version

The standard-library program `chai_verify_v1_increments.py` (123 checks, all passed; results in `chaiv1_increments_results.j` evaluates, from the primitive definitions, the statements that are new in this version of the paper, and the explicit constructions of its proofs on *every* binary function of the classes, not on samples:

- Section II: the three sections $\mathsf{NOT} = [+,+,0]$, $\mathsf{ANT} = [+,0,-]$, $\mathsf{DIF}(-1,\cdot) = [0,-,-] = -\mathsf{NOT}(-\cdot)$; that only the section at 0 is an involution; $\mathsf{NOT} \circ \mathsf{NOT} = [0,0,+]$; the least DIF-closed set containing $\{0,1\}$ is $\{-1,0,+1\}$ (iteration); $\mathsf{DIF}(-1,+1) = -1 \neq -2$.
- The printed Tables I, II and III against the definitions: all five binary tables and the three unary rows cell by cell; the five columns of Table II for the seven primitives; the numbers $3, 81, 12, 3888, 9, 6561, 27, 19683$ of Table III.
- Lemma 2 and Theorems 3–5: δ_p for the eight $p \neq 0$; all 6561 functions of $\mathscr{Z}_2$ from the five operations; all 19683 binary functions as $\mathsf{SYN}\big(F', \mathsf{AND}(e_0, F(0))\big)$ (Theorem 3(c)); all 81 odd binary functions from DIF alone with the terms k_c and s_p; all 3888 functions of $\mathscr{B}_2$ and 300 random functions of $\mathscr{B}_3$ from $\mathsf{SYN}, \mathsf{OR}, \mathsf{XOR}, \mathsf{AND}, \mathsf{NOT}$.
- Theorem 6: the criterion "$\mathsf{NOT} \in \Sigma$ and ($\mathsf{DIF} \in \Sigma$ or $\{\mathsf{SYN}, \mathsf{ANT}\} \subseteq \Sigma$)" agrees with all 127 rows of `classification_127.csv`, and 40 rows are complete; the relation ρ is preserved by $\mathsf{ANT}, \mathsf{OR}, \mathsf{XOR}, \mathsf{AND}, \mathsf{NOT}$ and not by SYN or DIF; the witnesses of the necessity proof, $\mathsf{DIF}(0,+1) = -1 \notin B$ and $(\mathsf{DIF}(+1,0), \mathsf{DIF}(0,+1)) = (+1,-1) \notin \rho$; the five closures with constants of Table IV of the paper against the shipped `results_consts_*.json` (size 19683 and last height).
- Sections III and VI: parity lemma and polarization at the nine pairs, the failure at $(2,1)$ on $\{-2,\ldots,2\}$, Φ an involution with one fixed point, the congruence of the fold (AND and ANT only), the clockwise rotation of the five tables, MAX, MIN as terms, the associativity failures $4, 16, 8, 8$, the Shannon/Post expansion on 500 random binary functions, the T9ID words of SYN and DIF, the order-8 group and the Burnside count 2862.

The large closures (the 6561, 3888 and 19683 binary functions as closures of the projections) and the closures with nullary constants are not recomputed here; they are those of the earlier verified runs (`results.json`, `results_consts_*.json`) for the same seven operations.

S19. Structure of Chai: the diagram of the earlier version

Figure S19 is the structure diagram of the earlier version of the paper, kept here because it shows the relations among the seven operations at a glance. Each arrow is a statement proved in the paper. Arrows from (a,b) to SYN and DIF: the pair $\Phi = (\mathsf{SYN}, \mathsf{DIF})$ is an involution (Theorem 2). Vertical arrows labelled MAG: folding the sum or the difference to its magnitude gives $\mathsf{OR} = \mathsf{MAG} \circ \mathsf{SYN}$ and $\mathsf{XOR} = \mathsf{MAG} \circ \mathsf{DIF}$. Arrows from OR and XOR to AND: the polarization $\mathsf{AND} = \mathsf{DIF}(\mathsf{OR}, \mathsf{XOR})$ (Theorem 1). Dashed double arrows: replacing the second argument b by $\mathsf{ANT}(b)$ exchanges $\mathsf{SYN} \leftrightarrow \mathsf{DIF}$ and $\mathsf{OR} \leftrightarrow \mathsf{XOR}$, since $\mathsf{SYN}(a,b) = \mathsf{DIF}(a, \mathsf{ANT}(b))$ and $\mathsf{OR}(a, \mathsf{ANT}(b)) = \mathsf{XOR}(a,b)$. The dotted rectangle is the zero-fixing clone $\mathscr{Z}$ ($f(0,\ldots,0) = 0$): the five binary operations stay inside it (Theorem 3a,b), while NOT is zero-breaking and, together with the five, gives all of $\mathscr{F}_1$ and $\mathscr{F}_2$ (Theorem 3c).

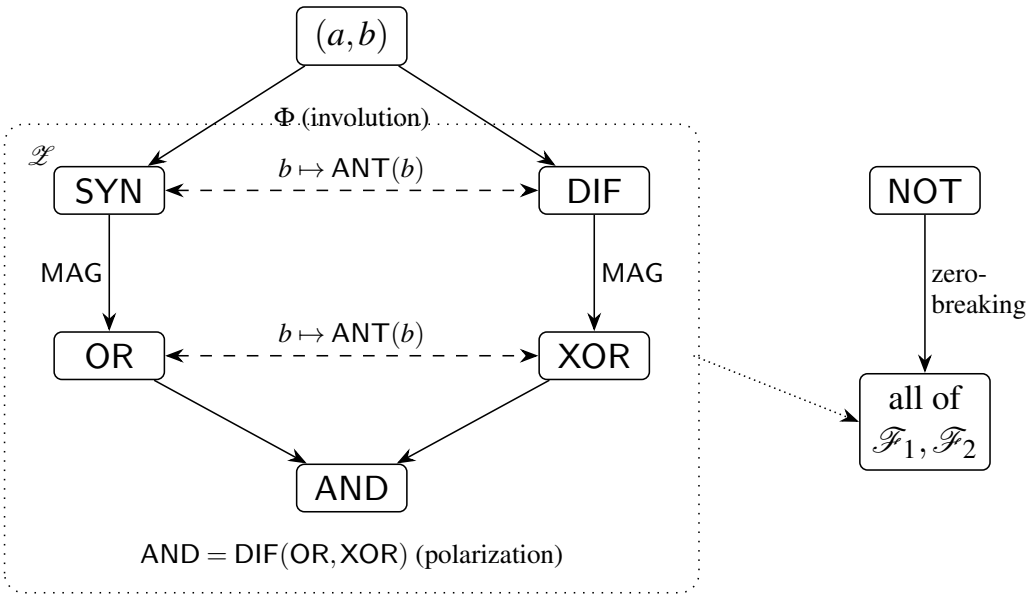


Figure S19. Structure of Chai. The five binary operations stay inside the zero-fixing clone $\mathscr{Z}$ (dotted; $f(0,\ldots,0) = 0$), and NOT leaves it. Dashed arrows: negation of the second argument.